CITYCARE GENERAL HOSPITAL

DRUG INTERACTION GUIDE STANDARD OPERATING PROCEDURE

Identification, Assessment & Management of Drug–Drug, Drug–Food & Drug–Disease Interactions

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1. PURPOSE & SCOPE

Drug interactions represent one of the most common and preventable causes of adverse drug events in hospital practice. This SOP establishes a systematic framework for the identification, risk stratification, clinical management, and documentation of all significant drug interactions encountered at CityHospital. It applies to drug–drug interactions (DDIs), drug–food interactions (DFIs), drug–nutrient interactions, and drug–disease interactions across all inpatient, outpatient, and emergency settings.

- Standardise the approach to identifying and managing drug interactions across all clinical settings.
- Prevent patient harm from undetected clinically significant drug interactions.
- Define clear escalation pathways when high-risk interactions are identified.
- Ensure all prescribing and dispensing staff are trained in interaction recognition and response.
- Maintain a continuously updated high-risk drug interaction reference within the EMR alert system.
- Support a culture of proactive medicines safety review during prescribing, dispensing, and administration.

NOTE: *This document provides clinical guidance and does not replace professional judgement. For complex cases or interactions not listed here, consult the clinical pharmacist or pharmacologist and reference the hospital's approved interaction database (Micromedex or equivalent).*

2. DEFINITIONS & INTERACTION CLASSIFICATION

Term	Definition
Drug–Drug Interaction (DDI)	A modification of the effect of one drug when another drug is co-administered, resulting in increased or decreased therapeutic effect or toxicity.
Drug–Food Interaction (DFI)	An alteration in drug absorption, metabolism, or effect caused by food, beverage, or specific dietary components consumed concurrently.
Drug–Nutrient Interaction	Interference between a drug and nutritional components (vitamins, minerals, enteral feeds) affecting absorption, efficacy, or nutritional status.
Drug–Disease Interaction	A situation where a drug's pharmacological action worsens an existing disease, comorbidity, or physiological state.
Pharmacokinetic Interaction	An interaction affecting drug absorption, distribution, metabolism (CYP450 enzymes), or excretion, altering plasma drug concentration.
Pharmacodynamic Interaction	An interaction where two drugs have additive, synergistic, or antagonistic effects at the receptor or physiological level, independent of concentration changes.
CYP450 Inhibitor	A drug that inhibits cytochrome P450 enzymes, reducing metabolism of co-administered drugs and raising their plasma levels.
CYP450 Inducer	A drug that induces CYP450 enzymes, accelerating metabolism of co-administered drugs and lowering their plasma levels.
Narrow Therapeutic Index (NTI)	Drugs where small changes in plasma concentration may cause toxicity or therapeutic failure. e.g., warfarin, digoxin, lithium, phenytoin.
QT Prolongation	Lengthening of cardiac repolarisation that can predispose to potentially fatal ventricular arrhythmia (Torsades de Pointes).
Serotonin Syndrome	Life-threatening reaction caused by excess serotonergic activity; triggered by drug combinations acting on serotonin pathways.
TDM	Therapeutic Drug Monitoring — measurement of drug serum levels to guide dosing and detect toxicity.

3. ROLES & RESPONSIBILITIES

Role	Department	Responsibilities
Clinical Pharmacist	Pharmacy	Daily prescription interaction screening, patient counselling, escalation of Class A interactions
Prescribing Physician	All Specialties	Check for interactions before prescribing; act on pharmacist alerts; document decisions
Ward Nurse	All Wards	Identify administration-level interactions; report patient symptoms consistent with interactions
Clinical Pharmacologist	Pharmacy / Clinical	Advise on complex or rare interactions; TDM interpretation; interaction database maintenance
Pharmacy Technician	Pharmacy	Flag EMR interaction alerts at dispensing stage; escalate to pharmacist
Chief Pharmacist	Pharmacy	Govern interaction alert library in EMR; ensure staff training compliance
IT / EMR Team	Information Systems	Maintain and update drug interaction decision support system in EMR

4. INTERACTION SEVERITY CLASSIFICATION SYSTEM

CityHospital uses a four-class severity system for drug interactions. The class determines the mandatory clinical response and documentation requirements.

Class	Severity	Definition	Required Action
A	CONTRAINDICATED	Combination poses an unacceptable risk of serious harm. Benefits almost never outweigh risks.	Do NOT prescribe. Pharmacist must contact prescriber immediately. If prescribed in error, reverse before dispensing. Document as clinical incident.
B	MAJOR	Interaction may be life-threatening or cause permanent damage. Avoid combination unless no alternative exists.	Pharmacist alerts prescriber within 30 minutes. Physician must document risk–benefit analysis in EMR. Increased monitoring mandatory. Consider alternatives.
C	MODERATE	Interaction may worsen patient condition or require treatment adjustment. Risk may be managed with monitoring.	Pharmacist flags at dispensing. Prescriber counselled. Monitoring parameters defined and documented. Patient educated.
D	MINOR	Limited clinical significance; interaction may be bothersome but unlikely to cause significant harm.	Noted in EMR. Patient counselled. No prescribing change usually required. Monitoring optional.

WARNING: Class A (CONTRAINDICATED) interactions identified by the EMR will trigger an IMMEDIATE hard-stop alert. The prescription cannot be dispensed until a pharmacist override code is entered with documented clinical justification — for use only in exceptional circumstances.

5. DRUG–DRUG INTERACTION (DDI) IDENTIFICATION WORKFLOW

5.1 At the Point of Prescribing — Physician

Step 1. Before prescribing any new drug, review the patient's complete active medication list in the EMR.

Step 2. The EMR clinical decision support system (CDSS) automatically screens all new prescriptions against the existing medication list and generates interaction alerts classified A to D.

Step 3. Physician reviews all Class A and B alerts immediately; documents clinical rationale in EMR before proceeding.

Step 4. For Class C alerts, physician acknowledges alert and documents monitoring plan.

Step 5. Class D alerts acknowledged; no mandatory action required but patient counsel recommended.

Step 6. If unsure about any interaction, contact the ward clinical pharmacist before prescribing.

5.2 At the Point of Dispensing — Pharmacist

Step 1. Pharmacist performs a second independent interaction check on all new prescriptions using the pharmacy dispensing system and the approved interaction database.

Step 2. Class A interaction identified: prescribing physician contacted immediately by phone; prescription held pending resolution.

Step 3. Class B interaction: physician contacted within 30 minutes; monitoring requirements agreed and documented in EMR.

Step 4. Class C interaction: interaction noted on dispensing label; patient counselled verbally and in writing.

Step 5. Pharmacist documents all interaction interventions in the EMR pharmacy intervention record.

5.3 At the Point of Administration — Nurse

Step 1. Before administering any drug, nurse reviews the patient's current medication administration record for administration-level interactions (e.g., timing separation, IV compatibility).

Step 2. If the patient reports new symptoms consistent with a drug interaction (palpitations, confusion, bleeding, nausea, hypotension), nurse withholds next dose and contacts physician immediately.

Step 3. IV drug compatibility checked using the hospital's approved IV compatibility reference before combining drugs in the same line or bag.

Step 4. Any suspected interaction-related adverse event documented in the EMR and reported as a clinical incident.

6. HIGH-RISK DRUG PAIRS — CLINICAL REFERENCE TABLES

6.1 Class A — Contraindicated Combinations

Drug 1	Drug 2	Mechanism	Risk
MAOIs (e.g., phenelzine)	SSRIs / SNRIs / Tramadol	Serotonin excess	SEROTONIN SYNDROME — potentially fatal
MAOIs	Pethidine / Meperidine	Serotonin + CNS toxicity	Hyperpyrexia, seizures, cardiovascular collapse
Warfarin	Miconazole (oral/topical)	CYP2C9 inhibition	Severe anticoagulation — fatal haemorrhage risk
Sotalol / Amiodarone	Haloperidol / Antipsychotics	Additive QT prolongation	Torsades de Pointes — fatal arrhythmia
Methotrexate (high-dose)	NSAIDs / Aspirin	Reduced renal MTX clearance	Methotrexate toxicity — bone marrow suppression
Simvastatin > 40mg	Clarithromycin / Itraconazole	CYP3A4 inhibition	Rhabdomyolysis — renal failure
Sildenafil / PDE5 inhibitors	Nitrates (any route)	Additive vasodilation	Severe hypotension — cardiovascular collapse
Linezolid	SSRIs / SNRIs / TCAs	MAO-A inhibition + serotonin load	Serotonin syndrome

6.2 Class B — Major Interactions Requiring Close Monitoring

Drug 1	Drug 2	Effect	Monitoring Required
Warfarin	Antibiotics (broad-spectrum)	Gut flora reduction — increased INR	INR within 48–72 hrs of antibiotic start
Digoxin	Amiodarone / Verapamil	Increased digoxin level — toxicity	Digoxin level; ECG; symptoms of toxicity
ACE inhibitors / ARBs	Potassium-sparing diuretics	Additive hyperkalaemia	Serum potassium within 1 week
Lithium	NSAIDs / Thiazide diuretics	Reduced lithium clearance — toxicity	Lithium level within 5–7 days of change
Metformin	IV Contrast media	Risk of lactic acidosis	Hold metformin 48 hrs post contrast; check eGFR
Cyclosporin	Statins (most)	CYP3A4/OATP1B1 inhibition	CK levels; myopathy symptoms
Heparin / LMWHs	Antiplatelet agents	Additive bleeding risk	Monitor for bleeding; check FBC
Phenytoin	Many drugs (inducer/inhibited)	Altered phenytoin levels	Phenytoin levels; clinical seizure monitoring

7. DRUG–FOOD & DRUG–NUTRIENT INTERACTIONS

Food can significantly alter drug absorption, distribution, and metabolism. All nursing and pharmacy staff must be familiar with the following key drug–food interactions and communicate these to patients at the time of prescribing and dispensing.

Drug	Food / Substance	Interaction Effect	Clinical Action
Warfarin	Vitamin K-rich foods (leafy greens, broccoli)	Antagonises anticoagulant effect — reduces INR	Consistent dietary intake; INR monitoring after dietary change
Warfarin	Grapefruit juice	CYP2C9 inhibition — raises warfarin level	Avoid grapefruit; check INR
Statins (simvastatin, atorvastatin)	Grapefruit / grapefruit juice	CYP3A4 inhibition — increases statin level	Avoid grapefruit; switch to pravastatin if unavoidable
MAOIs	Tyramine-rich foods (aged cheese, cured meats, red wine)	Hypertensive crisis — tyramine not metabolised	Strict dietary restriction; patient education essential
Tetracyclines / Fluoroquinolones	Dairy products, antacids, iron	Chelation — reduced drug absorption	Administer 2 hours before or 4 hours after food
Levodopa (Parkinson's)	High-protein meals	Amino acid competition — reduced CNS absorption	Administer 30 min before meals; reduce dietary protein timing
Oral bisphosphonates (alendronate)	Any food or drink except water	Reduced absorption; oesophageal irritation	Take with full glass of water; remain upright 30 min
Lithium	Low-sodium diet / excess sodium loss	Increased lithium reabsorption — toxicity	Consistent sodium intake; monitor with dietary changes
Enteral feeds	Phenytoin, ciprofloxacin, levothyroxine	Reduced drug absorption via tube binding	Hold feed 1–2 hrs before and after drug administration

NOTE: *Grapefruit, pomelo, and Seville orange contain furanocoumarins that irreversibly inhibit intestinal CYP3A4 for up to 72 hours. Even one glass can produce clinically significant interactions with over 85 drugs. Advise ALL affected patients to avoid these entirely during treatment.*

8. DRUG–DISEASE INTERACTIONS & CONTRAINDICATIONS

Certain drugs may exacerbate underlying disease states or be physiologically hazardous in patients with specific comorbidities. The following table provides a clinical reference for the most common and highest-risk drug–disease interactions encountered in hospital practice.

Drug / Drug Class	Comorbidity	Risk / Effect	Clinical Action
NSAIDs (ibuprofen, naproxen)	Peptic ulcer disease / GI bleeding history	Mucosal damage; rebleeding	Avoid; use paracetamol. If essential: add PPI
NSAIDs	Chronic kidney disease (eGFR < 30)	Renal vasoconstriction — AKI	Contraindicated; use paracetamol
NSAIDs	Heart failure	Sodium/water retention; worsening HF	Avoid; use paracetamol
Beta-blockers (non-selective)	Asthma / severe COPD	Bronchoconstriction	Contraindicated; use cardioselective if essential + close monitoring
Metformin	eGFR < 30 ml/min/1.73m²	Lactic acidosis accumulation	Contraindicated; switch to insulin
ACE inhibitors / ARBs	Bilateral renal artery stenosis	Precipitate AKI	Contraindicated; use alternative antihypertensive
Fluoroquinolones	Myasthenia gravis	Neuromuscular blockade worsening	Contraindicated; choose alternative antibiotic
Tricyclic antidepressants	Glaucoma (narrow-angle)	Anticholinergic effect — raised IOP	Contraindicated; consult ophthalmology
Opioids	Severe respiratory depression / COPD	Respiratory failure	Use with extreme caution; lowest dose; monitor SpO2
Antipsychotics (typical)	Parkinson's disease	Worsening motor symptoms — dopamine blockade	Avoid; use quetiapine or clozapine if antipsychotic essential
Thiazolidinediones (pioglitazone)	Heart failure	Fluid retention; worsening HF	Contraindicated in Class III–IV HF
Corticosteroids (systemic)	Diabetes mellitus	Hyperglycaemia; diabetic ketoacidosis	Monitor blood glucose 6-hourly; adjust insulin accordingly

9. PHARMACOKINETIC MONITORING & DOSE ADJUSTMENT

9.1 Therapeutic Drug Monitoring (TDM) — Indications

TDM is required for drugs with a narrow therapeutic index where sub-therapeutic levels cause treatment failure and toxic levels cause serious adverse effects. The following drugs require routine TDM at CityHospital:

Drug	Target Range	When to Sample	Interaction Impact on Level
Vancomycin	AUC/MIC 400–600 mg·h/L	Trough before 4th dose; AUC-guided	Aminoglycosides — additive nephrotoxicity
Gentamicin (once-daily)	Trough < 1 mg/L	Trough at 18–24 hours post dose	NSAIDs, loop diuretics — raised levels
Digoxin	0.5–0.9 nmol/L (HF)	Trough at least 6 hrs post dose	Amiodarone, verapamil — raises level 2x
Phenytoin	40–80 µmol/L (total)	Trough before morning dose	Many CYP inducers/inhibitors — unstable level
Lithium	0.6–1.0 mmol/L (maintenance)	12 hrs post last dose	NSAIDs, thiazides — raises level
Ciclosporin	Varies by indication (50–400 ng/ml)	Trough (C0) before morning dose	Macrolides, azoles, St John's Wort
Methotrexate (high-dose)	Levels per protocol	24, 48, 72 hr post infusion	NSAIDs — delayed clearance; toxicity
Theophylline	10–20 mg/L	Trough before dose	Fluoroquinolones, macrolides — raise level

9.2 CYP450 Key Interactions Reference

Enzyme	Major Inhibitors (RAISE levels of substrates)	Major Inducers (LOWER levels of substrates)	Key Substrates
CYP3A4	Clarithromycin, ketoconazole, ritonavir, grapefruit	Rifampicin, phenytoin, carbamazepine, St John's Wort	Simvastatin, cyclosporin, midazolam, amlodipine
CYP2C9	Fluconazole, miconazole, amiodarone	Rifampicin, carbamazepine	Warfarin, phenytoin, NSAIDs (celecoxib)
CYP2C19	Omeprazole, fluoxetine, fluvoxamine	Rifampicin, carbamazepine	Clopidogrel (pro-drug), diazepam, PPIs
CYP2D6	Fluoxetine, paroxetine, haloperidol	Dexamethasone (minor)	Codeine, tramadol, metoprolol, TCAs

10. REPORTING, AUDIT & BEST PRACTICES

10.1 Interaction-Related Adverse Event Reporting

- Any patient harm caused by a drug interaction must be reported as a clinical incident in the EMR within 24 hours.
- Class A or B interaction errors (where contraindicated combination was prescribed and reached the patient) constitute a Level 3–5 incident requiring Root Cause Analysis.
- Suspected drug interactions causing unexpected clinical deterioration: pharmacist and physician complete a joint ADR report and submit to the national pharmacovigilance authority.
- All interaction-related near misses and incidents reviewed at the monthly Medication Safety Committee meeting.
- Learning from interaction incidents shared via safety bulletin to all clinical staff within 2 weeks of RCA completion.

10.2 EMR Drug Interaction Alert System Governance

- Drug interaction decision support (CDSS) module reviewed and updated quarterly by the clinical pharmacologist using the latest evidence.
- Alert fatigue monitoring: if override rate for any Class A or B alert exceeds 20%, alert text and trigger criteria reviewed for clinical relevance.
- New drug added to formulary: interaction profile reviewed by pharmacist before EMR activation; alerts configured within 24 hours of drug going live.
- All alert overrides for Class A interactions logged and audited monthly by Chief Pharmacist.

10.3 Drug Interaction Audit Programme

Audit Indicator	Target	Frequency	Responsible
Class A interaction prescriptions dispensed without authorised override	Zero	Weekly	Chief Pharmacist
Class B interactions with documented monitoring plan in EMR	100%	Monthly random sample	Clinical Pharmacist
Pharmacist intervention rate (interactions identified at dispensing)	Tracked for trend	Monthly	Pharmacy Manager
TDM ordered for all applicable NTI drugs	100%	Monthly	Clinical Pharmacist
Drug–food interaction counselling documented at discharge	Greater than 90%	Monthly	Ward Pharmacist

10.4 Patient-Facing Responsibilities

- Patients prescribed warfarin, lithium, digoxin, ciclosporin, or any NTI drug receive a personalised drug interaction card listing key food, OTC drug, and lifestyle interactions to avoid.
- Patients counselled verbally and in writing about interactions with alcohol, grapefruit, and herbal supplements (St John's Wort, evening primrose, ginkgo, garlic) for all relevant medications.
- Patients encouraged to carry an up-to-date medication list and present it to any treating clinician, dentist, or community pharmacist.
- Medication review appointment offered to all patients on 5 or more long-term medications at every discharge or OPD visit.

10.5 Best Practices for Safe Prescribing

- Never dismiss an EMR interaction alert without reading it; even Class C and D alerts may become significant in a vulnerable patient.
- Consider the whole patient: interactions become more dangerous with increasing age, renal impairment, hepatic dysfunction, low body weight, or polypharmacy.
- When starting a new drug in a patient on warfarin or lithium: plan monitoring proactively before the new drug is administered, not after.
- Use the pharmacist as a clinical partner: the ward pharmacist is trained to identify and resolve interactions and should be consulted for all complex patients.
- Annual interaction awareness refresher training is mandatory for all prescribers, pharmacists, and senior nurses.